

Q. Technologies Covered

- Cucumber BDD
- Selenium WebDriver
- TestNG
- Maven
- Page Object Model (POM)
- Data Driven

Framework Scenario 1 – E-Commerce End-to-End Flow

Use Case Automate complete e-commerce workflow.

Requirements - 1. Launch browser 2. Login using data from Excel 3. Search product 4. Apply filter 5. Add multiple products to cart 6. Remove one product 7. Validate total amount 8. Proceed to checkout 9. Logout 10. Capture screenshot on failure

Tasks for Learners

Task 1- Create Feature File with: Scenario Outline

Task 2 - Implement: Explicit Waits

Task 3 Read: Username/password

CODES:

CartPage.java

```
package pages;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.WebDriver;
```

```
public class CartPage {
```

```
    WebDriver driver;
```

```
    By removeBtn = By.id("remove-sauce-labs-bike-light");
```

```
    By checkoutBtn = By.id("checkout");
```

```
    public CartPage(WebDriver driver) {
```

```
        this.driver = driver;
```

```
    }
```

```
    public void removeProduct() {
```

```
        driver.findElement(removeBtn).click();
```

```
    }
```

```
    public void checkout() {
```

```
        driver.findElement(checkoutBtn).click();
```

```
    }
```

```
}
```

CheckoutPage.java

```
package pages;
```

```
public class CheckoutPage {
```

```
}
```

LoginPage.java

```
package pages;
```

```
import org.openqa.selenium.By;
```

```
import org.openqa.selenium.WebDriver;
```

```
public class LoginPage {
```

```
    WebDriver driver;
```

```
    By username = By.id("user-name");
```

```
    By password = By.id("password");
```

```
    By loginBtn = By.id("login-button");
```

```
    public LoginPage(WebDriver driver) {
```

```
        this.driver = driver;
```

```
    }
```

```
    public void login(String user, String pass) {
```

```
        driver.findElement(username).sendKeys(user);
```

```
        driver.findElement(password).sendKeys(pass);
```

```
        driver.findElement(loginBtn).click();
```

```
    }
```

```
}
```

ProductPage.java

```
package pages;
```

```

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;

public class ProductPage {

    WebDriver driver;

    By addBag = By.id("add-to-cart-sauce-labs-backpack");

    By addBike = By.id("add-to-cart-sauce-labs-bike-light");

    By cart = By.className("shopping_cart_link");

    public ProductPage(WebDriver driver) {

        this.driver = driver;
    }

    public void addProducts() {

        driver.findElement(addBag).click();

        driver.findElement(addBike).click();
    }

    public void openCart() {

        driver.findElement(cart).click();
    }
}

package runner;

import io.cucumber.testng.AbstractTestNGCucumberTests;

import io.cucumber.testng.CucumberOptions;

@CucumberOptions(

    features = "src/test/resources/ecommerce.feature",

    glue = "stepdefinitions",

    plugin = {

        "pretty",

```

```

"html:target/cucumber-report.html"

},

monochrome = true

)

public class TestRunner extends AbstractTestNGCucumberTests {

}

package stepdefinitions;

import io.cucumber.java.en.*;

import pages.CartPage; import
pages.LoginPage; import
pages.ProductPage; import
utilities.BaseClass;
import utilities.ExcelUtility;

public class EcommerceSteps extends BaseClass {

    LoginPage lp;

    ProductPage pp;

    CartPage cp;

    @Given("User launches browser")
    public void user_launches_browser() {

        setup();
    }

    @When("User enters {string} and {string}")
    public void user_enters_and(String username, String password) {

        lp = new LoginPage(driver);

        String user = ExcelUtility.getData(1, 0);

        String pass = ExcelUtility.getData(1, 1);
        lp.login(user, pass);
    }
}

```

```

@And("User searches product {string}")
public void user_searches_product(String product) {

    System.out.println("Product Searched");
}

@And("User applies filter")
public void user_applies_filter() {

    System.out.println("Filter Applied");
}

@And("User adds products to cart")
public void user_adds_products_to_cart() {

    pp = new ProductPage(driver);

    pp.addProducts();

    pp.openCart();
}

@And("User removes one product")
public void user_removes_one_product() {

    cp = new CartPage(driver);

    cp.removeProduct();
}

@Then("Validate total amount")
public void validate_total_amount() {

    System.out.println("Amount Validated");
}

@When("User proceeds to checkout")
public void user_proceeds_to_checkout() {

    cp.checkout();
}

@And("User logs out")
public void user_logs_out() {
closeBrowser();
}
}

```

```
package utilities;

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

public class BaseClass {

    public static WebDriver driver;

    public void setup() {

        driver = new ChromeDriver();

        driver.manage().window().maximize();

        driver.get("https://www.saucedemo.com/");
    }

    public void closeBrowser() {

        driver.quit();
    }
}
```

```
package utilities;

import java.io.FileInputStream;

import org.apache.poi.ss.usermodel.Cell; import
org.apache.poi.xssf.usermodel.XSSFSheet; import
org.apache.poi.xssf.usermodel.XSSFWorkbook; public
class ExcelUtility {

    public static String getData(int row, int cell) {
```

```
String value = "";

try {

    FileInputStream fis = new FileInputStream(
        "src/test/resources/TestData.xlsx");

    XSSFWorkbook workbook = new XSSFWorkbook(fis);

    XSSFSheet sheet = workbook.getSheet("Login");

    Cell data = sheet.getRow(row).getCell(cell);

    value = data.toString();

    workbook.close();

} catch (Exception e) {

    System.out.println(e);
}

return value;
}

package utilities;

import java.io.File;
```

```
import org.apache.commons.io.FileUtils;
```

```
import org.openqa.selenium.OutputType;
```

```
import org.openqa.selenium.TakesScreenshot;
```

```
public class ScreenshotUtility {
```

```
    public static void captureScreenshot(String name) {
```

```
        try
```

```
{
```

```
    File src = ((TakesScreenshot) BaseClass.driver)
```

```
        .getScreenshotAs(OutputType.FILE);
```

```
    File dest = new File("Screenshots/" + name + ".png");
```

```
    FileUtils.copyFile(src, dest);
```

```
    } catch (Exception e) {
```

```
        System.out.println(e);
```

```
    }
```

```
}
```

```
}
```

```
package utilities;
```

```
import java.time.Duration;
```

```
import org.openqa.selenium.WebDriver; import
org.openqa.selenium.WebElement; import
org.openqa.selenium.support.ui.ExpectedConditions; import
org.openqa.selenium.support.ui.WebDriverWait;

public class WaitUtility {

    public static void waitForElement(WebDriver driver, WebElement element) {

        WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));

        wait.until(ExpectedConditions.visibilityOf(element));
    }
}
```

Feature: E-Commerce End to End Flow

Scenario Outline: Complete shopping workflow

Given User launches browser

When User enters "<username>" and "<password>"

And User searches product "Laptop"

And User applies filter

And User adds products to cart

And User removes one product

Then Validate total amount

When User proceeds to checkout

And User logs out

Examples:

| username | password |
| standard_user | secret_sauce |

```
<project xmlns="http://maven.apache.org/POM/4.0.0"  
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
    https://maven.apache.org/xsd/maven4.0.0.xsd">
```

```
  <modelVersion>4.0.0</modelVersion>
```

```
  <groupId>com.Assignment_27</groupId>
```

```
  <artifactId>EcommerceFramework_Asmt_27</artifactId>
```

```
  <version>0.0.1-SNAPSHOT</version>
```

```
  <dependencies>
```

```
    <!-- Selenium -->
```

```
    <dependency>
```

```
      <groupId>org.seleniumhq.selenium</groupId>
```

```
      <artifactId>selenium-java</artifactId>
```

```
      <version>4.21.0</version>
```

```
    </dependency>
```

```
    <!-- Cucumber -->
```

```
    <dependency>
```

```
      <groupId>io.cucumber</groupId>
```

```
      <artifactId>cucumber-java</artifactId>
```

```
      <version>7.18.0</version>
```

</dependency>

<dependency>

 <groupId>io.cucumber</groupId>

 <artifactId>cucumber-testng</artifactId>

 <version>7.18.0</version>

</dependency>

<!-- TestNG -->

<dependency>

 <groupId>org.testng</groupId>

 <artifactId>testng</artifactId>

 <version>7.10.2</version>

 <scope>test</scope>

</dependency>

<!-- Apache POI -->

<dependency>

 <groupId>org.apache.poi</groupId>

 <artifactId>poi-ooxml</artifactId>

 <version>5.2.5</version>

</dependency>

</dependencies>

</project>

1 PASSED		
100% passed 1 executed	5 minutes ago last run	1 minute 35 seconds duration
 Windows 11	 OpenTK 64-bit Server VM 17.04-8	 transfer-jun-7-18.0
Search with text or image You can search with plain text or upload an image to filter the output		
»  file:///C:/Users/91361/Desktop/workshop%20exercise%20network_files/27for%20test%20resources/communication		